

PROJECT TITLE : AI FAQ ASSISTANT

PROJECT DOCUMENTATION PHASE

DOCUMENTATION OVERVIEW:

The documentation phase focused on preparing a complete and systematic record of the **AI FAQ Assistant** project. It describes the project requirements, system architecture, database structure, major features, development process, API operations, security mechanisms, and testing procedures. This documentation provides a clear understanding of how the system is designed, developed, and operated.

REQUIREMENT DOCUMENTATION:

The required software, hardware, and technologies were documented to define the environment needed for project development and execution. The system uses **Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, and Google Gemini SDK** to support authentication, database management, FAQ operations, and AI-powered content generation.

SYSTEM DESIGN DOCUMENTATION:

The system design section documents the layered architecture of the application, including the API interface, Express server, authentication middleware, business logic modules, database interface, and Google Gemini integration. The database entities such as **User, FAQ, and Category**, along with their relationships, are also documented.

FUNCTIONAL DOCUMENTATION:

The major functionalities of the system are documented, including **user authentication, JWT-based authorization, FAQ creation and management, category management, keyword and semantic search, AI FAQ generation, password hashing, validation, and centralized error handling**. The roles and permissions of Admin, Content Creator, Authenticated User, and Public User are also described.

API AND TESTING DOCUMENTATION:

The important API endpoints for registration, login, FAQ creation, FAQ search, and AI-based FAQ generation are documented along with their request methods and authentication requirements. Testing procedures using **Postman or Thunder Client** are also included to verify the proper functioning of the developed APIs.

DOCUMENTATION OUTCOME:

The completed documentation provides a structured and professional reference for the entire AI FAQ Assistant project. It helps users and developers understand the system requirements, architecture, functionalities, implementation, security, API operations, and testing process, making the project easier to understand, maintain, and further develop.